# Gate B v2 and metrics: change log

What changed, why, what the re-run showed, and what the dashboard now shows.

**Where this stands.** All six phases are done on the branch gate-b-v2-and-metrics-consolidation. It is not pushed, as you asked, and not merged, so this shows on port 8101 only.

**Cost and cash now come only from the card.** The backtest always ran at your 0bp. The 30bp was added afterwards, to Gate v2 and a "net" column, by a rule in the brief that I applied over your Gate A ruling. That is removed, along with every other cost or cash number in the code.

**Clenow's latest run (20261105-152026-31c4) now reads PASS on Gate v2, at your 0bp.** Bootstrap P 0.054 clears the eased 0.10, but would miss the original 0.05. Factor alpha is amber at t 1.17. At 0bp nothing here tests what trading 10.3 times capital a year would cost: the sweep below does that.

## Your rulings

| You said | What I did |
|---|---|
| **Use 0bp as set at Gate A. Remove all hard-coding of cost and cash** | Card only: the gate, metrics, page and Excel use costs.spread_bps and universe.cash_asset. Details below |
| **Approve, but make them easier** | Bootstrap P eased from 0.05 to 0.10, and sub-period to 2 of 4 windows, none below -1.0. Pass at a true excess Sharpe of 0.5 rose from 42.6% to 57.1%. Alpha stays at t 2.0: any lower breaks the 5% false-pass target |
| **Benchmark Sharpe** | Where the deflated Sharpe bar starts. Dormant while the test is out |
| **Remove the deflated Sharpe for now** | Out of the gate, off in config. **So v2 no longer tests for selection luck**. The results table still reports it as a figure |
| **INCOMPLETE is not the right way** | Status is PASS or FAIL. A test with nothing to run on shows as not run and is listed beside the status |
| **Good to proceed without those values** | AUM, mandate limits, the book and product costs stay blank. Their tests show as not run |
| **Archive, but keep the tearsheet for Excel** | **Nothing archived:** the Excel calculation needs both tearsheet files, and the momentum rotation script produces that workbook |
| **Hold the push** | Not pushed |

## Cost and cash: what was removed

| Where | Before | Now |
|---|---|---|
| **Gate v2 and the "net" column** | Charged a 30bp house cost over your 0bp | The run's returns as they are, at the card's cost. Gross shows beside net only if that cost is not zero |
| **config/costs.yaml** | house_round_trip_bp: 30 | Only the appendix sweep grid. It never reaches a headline or a gate |
| **config/fund.yaml** | A 6% fallback I added yesterday | Removed. Only the appendix cash sweep. A card naming no rate gets Sharpe over zero, and the page says so |
| **Clenow script** | A dedicated 30bp comparator and two rows at 30bp and 100bp. Grids written in code | Removed. Grids read from config. A breakeven row replaces them |
| **Rule text: CLAUDE.md, Gate A requirements, card drafting, universes, the model's prompts** | "30bp is non-negotiable" | The card states a cost and a human confirms it at Gate A |
| **Code defaults** | Cost 5bp and 10bp in the engines and schema. Cash 6% in the pipeline, runner, cross-check and deciles, and 5% in the Excel tearsheet | No defaults. A missing value is an error, or comes from the card |
| **Completeness and Gate A checks** | A cost under 20bp counted as a fault | The check is that the card states one. Zero is a stated cost, and cost is always flagged for a human |
| **Momentum rotation script** | 30bp, and 5% on the Excel sheet | It has no card, so you pass both when you run it |
| **Simulations** | 6% and 30bp written in | In config/calibration.yaml, read by the simulations only. No strategy run reads them |

A test now fails if a number like these is written back into live code.

## Commits

| Commit | What | Why |
|---|---|---|
| 925289c | Freeze Gate B v1: version constants, manifest and workbook fields, baseline for every past run | So every later result can be compared with what v1 really said. |
| 5139c54 | One metrics module: every dashboard, workbook, report and gate reads ros/metrics/core.py | The same metric name gave different numbers depending on where you looked. |
| fab1b88 | Gate v2: hard blocks, evidence scorecard, PASS/FAIL/INCOMPLETE, mandate file, card fields | Built so it could be simulated, before any threshold was chosen. |
| c471338 | Calibrate Gate B by simulation: 8,000 strategies, real-data references, power curves and a proposal | So thresholds come from simulation, not from any one strategy. |
| b26130c | Fixes from the dashboard check: first-day trades count in turnover, older runs label their old Sharpe, runs stop copying core metrics | The dashboard check found these. |
| a73e1e9 | Change log for the metrics consolidation and Gate v2 work, with the re-run and dashboard check | You asked for a record of the changes. |
| fb6bdc7 | Approve easier Gate v2 thresholds, deflated Sharpe out, missing inputs show as not run | Your rulings, committed before any named strategy was scored on them. |
| 0d5a58f | Wire Gate v2 into Lightyear: scored on every run, new Gate B tab and workbook sheet, past runs re-scored | So every run is scored on v2, on the page and in Excel. |
| 6818b7f | Page, workbook and tests name where the cash rate came from | Superseded by the card-only rule below. |
| 0b3ea3f | Reference document updated to v2: one formula per metric, the gate, its calibration, changes from v1; end-to-end gate tests | The document updated to v2. |
| 684ef7b | Change log brought up to date: rulings, Gate v2 wired, v1 against v2 for every run, the Clenow comparison | You asked for a record of the changes. |
| ded8616 | A run's cost and cash come only from its card: no 30bp, 6% or other number written into code | **Your ruling: the cost set at Gate A is the cost charged.** |
| 1a76473 | A value with no finite answer shows blank instead of breaking the Results page | My new breakeven row broke the page at 0bp. |

83 files changed, 5238 insertions(+), 479 deletions(-), compared with the tag gate-v1. Tests: pending on the final code (570 passed before this work). ruff: no new errors against the original code.

## Metric definitions: before and after

| Metric | Before | Now |
|---|---|---|
| **Sharpe** | Four formulas: (CAGR - 6%) / vol on the tile, (CAGR - compounded cash) / vol in tables, (mean x 252 - 6%) / vol in Excel, mean excess / stdev in the deflated Sharpe | One: mean(r - c) / stdev(r - c) x sqrt(252) |
| **Max drawdown** | Positive in some places, negative in others | Daily values, positive |
| **Drawdown duration** | Calendar days | Trading days |
| **Tracking error** | Sample in one path, population in another | Sample stdev of daily gaps |
| **Up / down capture** | Daily means in one path, monthly in another | **Monthly means**. My choice: your brief named no frequency |
| **95% CVaR** | Rolling 21-day in one path, daily and negative in another | **Daily, positive loss**. My choice of frequency |
| **Turnover** | Mean x 252 in one path, sum / years in another | Sum / years |
| **Bootstrap** | Geometric Sharpe | The one Sharpe, same settings, plus a paired drawdown gap as evidence only |

## Added

| What | Where |
|---|---|
| **Core performance table: the same 16 metrics on every run, including Sortino and Calmar** | Results tab; Core metrics sheet in Excel |
| **Gate v2: hard blocks, scorecard, PASS or FAIL. v1 folded below for comparison** | Gate B tab; Gate v2 sheet in Excel |
| **Capacity. Code done, shows as not run until you supply AUM** | Results tab, Gate B tab |
| **Post-publication metrics** | Results tab, Excel, scorecard |
| **Diagnostics section for rows tied to one strategy's rules** | Results tab, Excel Validation sheet |
| **gate_version and metrics_version** | manifest.json, workbook header, Gate B tab |
| **Card fields: n_configs_selected_from, n_sensitivities, rules_fixed_in_print, fits_parameters** | ros/cards/schema.py |

## Cut and kept

| Item | Status |
|---|---|
| **Mean drawdown, vol consistency, hit rate, monthly CVaR, the geometric and duplicate Sharpes** | **Cut** from the pipeline report and metrics modules |
| **Sterling, Treynor, Omega, Gain to Pain, tail ratio, adjusted beta, Jensen's alpha, extreme capture and the rest of your cut list** | **Off Lightyear, still in the tearsheet**, because you need the Excel calculation. Its numbers can differ from the dashboard's |
| **Geometric Sharpe** | **Kept in one place**, so a paper's printed Sharpe compares on its own definition |
| **validate/independent.py** | **Kept separate on purpose:** the clean-room cross-check imports nothing from ros |
| **universal_backtester/validation.py** | **Not consolidated.** Its own bootstrap and deflated Sharpe remain, but no Lightyear path or gate uses them |

## Gate v2 and the calibration

| Share passing | True Sharpe 0 | 0.5 | 0.75 | 1.0 | Factor only | Best of 20 | Best of 100 |
|---|---|---|---|---|---|---|---|
| **Status PASS (hard blocks)** | 6.9% | 63.2% | 87.4% | 95.7% | 73.4% | 72.5% | 97.3% |
| **PASS and all green** | 4.5% | 57.1% | 84.7% | 95.3% | 2.6% | 59.6% | 95.9% |

> **What the rulings cost.** PASS and all green meets the targets for no-skill and factor-only strategies. **Status PASS alone does not:** 73% of factor-only strategies and 6.9% of no-skill ones read PASS, because the scorecard does not block. With the deflated Sharpe out, 96% of the best of 100 random strategies get PASS and all green. And a card at 0bp is charged nothing, so PASS there says nothing about trading costs.

Thresholds came from simulations and were committed (fb6bdc7) before Clenow was scored on them. To be open about it: I had already seen Clenow's zero-cost P of 0.054 when I eased the bar past it. Formulas and curves: financial-metrics-and-gate-b-v2.pdf. gate-b-v2-calibration.pdf is the earlier proposal.

## Clenow's latest run: v1 against v2

| Test | Gate v1 | Gate v2, at the card's 0bp |
|---|---|---|
| **Status** | **FAIL**, 1 of 5 blocking | **PASS** |
| **Edge over the benchmark** | Not tested | 7.05%: PASS |
| **Paired bootstrap P** | 0.054: fail at 0.05 | 0.054: PASS at 0.10 |
| **Worst window** | -0.24 absolute Sharpe: fail at 0.30 | Sharpe gap 0.11, 4 of 4 windows ahead: green |
| **Factor alpha t** | 1.52 on 2 factors: fail at 2.0 | 1.17 on 7 factors: **amber**, does not block |
| **Deflated Sharpe** | 0.00: fail | Out of the gate |
| **After publication: edge** | Not tested | 5.51%: green |
| **Data integrity** | Not tested | PASS |
| **Mandate, capacity, book** | Mandate passed on the card's flag; no book | Not run: no values supplied |

| Appendix sweep: round-trip cost | 0bp | 10bp | 20bp | 30bp | 50bp | 75bp | 100bp |
|---|---|---|---|---|---|---|---|
| **CAGR** | 19.80% | 19.25% | 18.52% | 17.95% | 16.74% | 15.16% | 13.72% |
| **Edge over the index** | 6.96% | 6.42% | 5.68% | 5.12% | 3.90% | 2.33% | 0.88% |

| On the page now | Strategy, at the card's cost | NIFTY 500 |
|---|---|---|
| **CAGR** | 19.80% | 12.83% |
| **Sharpe** | 0.83 | 0.49 |
| **Sortino** | 1.10 | 0.65 |
| **Calmar** | 0.88 | 0.34 |
| **Maximum drawdown** | 22.47% | 38.30% |
| **Drawdown duration (trading days)** | 547 | 490 |
| **Information ratio** | 0.49 | – |

Daily returns, trades and holdings are byte-identical to the earlier runs of this card. Turnover 10.33 times capital a year. Breakeven is beyond the sweep.

## Every past run: v1 against v2

All 14 runs failed v1. Under v2, each at its own card's cost, 9 pass and 5 fail. Most of the passes are runs of the same Clenow card. Full detail: docs/gate_b/v1_vs_v2.csv.

| Run | Strategy | v1 | v2 | Edge | P | Why it changed |
|---|---|---|---|---|---|---|
| 20261006-181257-0394 | Padhy 130/30 | FAIL | **FAIL** | 6.60% | 0.468 | Still fails: bootstrap P above 0.10 |
| 20261006-lee-swaminathan-1998 | Lee-Swaminathan | FAIL | **FAIL** | -11.79% | 0.941 | Still fails: bootstrap P above 0.10 |
| 20261013-161034-6649 | 1_s2_0 (s0970389623000538) | FAIL | **PASS** | 6.86% | 0.047 | Clears every hard block that could be run. Alpha (t 0.89) no longer blocks, and the deflated Sharpe is out |
| 20261016-102923-bdc1 | Clenow | FAIL | **FAIL** | 9.05% | 0.189 | Still fails: bootstrap P above 0.10 |
| 20261021-104144-801e | Clenow | FAIL | **FAIL** | 9.05% | 0.189 | Still fails: bootstrap P above 0.10 |
| 20261022-153434-39aa | Momentum 50 rebuild | FAIL | **FAIL** | 4.25% | 0.550 | Still fails: bootstrap P above 0.10 |
| 20261030-124039-5511 | Clenow | FAIL | **PASS** | 8.19% | 0.032 | Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out |

| Run | Strategy | v1 | v2 | Edge | P | Why it changed |
|---|---|---|---|---|---|---|
| 20261030-171327-0aa8 | Clenow | FAIL | **PASS** | 8.19% | 0.032 | Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out |
| 20261030-173539-8fd8 | Clenow | FAIL | **PASS** | 8.19% | 0.032 | Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out |
| 20261030-183103-3639 | Clenow | FAIL | **PASS** | 8.19% | 0.032 | Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out |
| 20261030-191604-3399 | Clenow | FAIL | **PASS** | 8.19% | 0.032 | Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out |
| 20261102-105054-0427 | Clenow | FAIL | **PASS** | 7.05% | 0.054 | Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out |
| 20261105-132328-ab35 | Clenow | FAIL | **PASS** | 7.05% | 0.054 | Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out |
| 20261105-152026-31c4 | Clenow | FAIL | **PASS** | 7.05% | 0.054 | Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out |

## Dashboard check

I opened the re-run in a real browser on the final code: **47 of 47** checks pass, with no script errors. Excel, after recalculation, matches the page on CAGR, volatility, Sharpe, drawdown, Sortino and Calmar. Run it again with scripts/gate_v2/check_dashboard.py.

## Problems found and fixed

| Problem | Cause | Fix |
|---|---|---|
| **The gate charged 30bp on a 0bp card** | A rule in the brief, which I applied over your Gate A ruling instead of raising the conflict | Card only |
| **The Results page returned a server error at 0bp** | **My bug.** Breakeven is infinite when the edge never reaches zero, and the page cannot carry that | Shows blank, and no value can do this again |
| **Turnover read 10.33 and 10.26** | **My bug.** The first day's purchases were dropped | Counted on the first day |
| **Older runs said "Sharpe" but showed the old 6% number** | My relabel | Labelled as the old formula, with a note to press Rebuild |
| **7 rule-specific rows sat in Costs & execution** | Written during the run | Lightyear moves them to Diagnostics |
| **A factor equal to the strategy gave alpha t 17.6** | A series regressed on itself | An identical factor is left out, and the row says so |

## Still open

| Item | Why it matters |
|---|---|
| Claude's written text on this run still says "the 30bp house floor", in its strengths, weaknesses and notes | Written before the instruction changed. Future runs are told not to. Nothing enforces prose, as with "Evidence supports" |
| Mandate, capacity and book tests show not run | A PASS has not been tested on these. Filling in config/fund.yaml and config/mandate.yaml switches them on |
| The Gate B brief and "Evidence supports" are written from the v1 checklist | They can disagree with the v2 status above them. Labelled on the page |
| Each run can still add its own rows | "Rolling one-year behind" can show two values. Runs are told not to, but nothing enforces it |
| Older runs show v2 only after Rebuild | The table above scores them all, but their pages need Rebuild |
| config/calibration.yaml still holds 6% and 30bp | For the simulations only. No strategy run reads it |
| Your usual Lightyear on port 8100 | It is on the old branch, and was not running when I last checked |